

IM M V POPOVA, Russian Federation

WMO: 206670

Lat: 73.333N

Lon: 70.066E

Elev: 18

StdP: 14.69

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.0	-31.4	-39.6	0.6	-34.5	-36.0	0.7	-31.2	36.2	7.1	32.1	12.5	6.8	120	1.081

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	58.2	54.5	53.7	50.9	50.2	48.2	54.9	57.8	51.2	53.4	48.2	50.0	10.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.0	59.7	56.7	49.7	52.8	52.7	46.8	47.3	49.4	23.2	57.9	21.0	53.4	19.3	50.0	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	28.7	25.7	-37.0	64.6	5.2	8.8	-40.7	71.0	-43.8	76.1	-46.7	81.1	-50.4	87.5		
(5)			-37.1	58.7	5.1	6.1	-40.8	63.0	-43.8	66.6	-46.7	70.0	-50.5	74.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	-6.4	-6.4	-4.1	5.0	19.9	32.7	42.9	42.6	37.1	24.8	7.8	-1.3
	DBStd	21.48	14.53	15.97	14.13	11.11	7.73	4.30	6.52	3.99	4.96	9.61	11.83	15.07
	HDD50	12308	1748	1579	1676	1349	934	520	242	235	387	782	1265	1591
	HDD65	17756	2213	1999	2141	1799	1399	970	686	695	836	1247	1715	2056
	CDD50	28	0	0	0	0	0	1	21	5	1	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.5	12.5	12.0	12.3	12.0	12.7	12.3	11.2	12.5	12.5	13.5	12.8	13.4
Precipitation	PrecAvg	8.1	0.4	0.4	0.4	0.4	0.4	0.7	0.7	1.2	1.1	0.7	0.6	0.7
	PrecMax	19.1	0.9	1.5	0.8	1.0	1.0	4.2	1.8	5.0	2.9	1.9	1.9	1.8
	PrecMin	2.4	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.3	0.2	0.0	0.1
	PrecStd	4.1	0.3	0.4	0.2	0.3	0.3	0.8	0.5	1.0	0.7	0.4	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.7	29.7	30.3	32.9	34.7	55.7	71.0	60.1	54.3	40.7	31.8	30.3
		MCWB	27.8	29.1	29.8	31.9	33.4	50.6	62.4	56.2	51.8	39.2	31.5	29.4
	2%	DB	24.7	26.6	26.2	30.2	33.1	45.9	61.7	54.6	49.0	38.8	29.7	28.3
		MCWB	23.9	25.9	25.6	29.4	32.2	43.6	57.1	51.7	47.8	37.8	29.2	27.7
	5%	DB	21.1	23.1	22.7	26.9	32.0	40.4	56.7	51.4	46.0	36.8	27.7	25.9
		MCWB	20.3	22.6	22.2	26.2	31.4	39.4	53.1	49.9	44.9	35.7	27.0	25.3
	10%	DB	16.2	18.6	18.7	23.3	30.5	37.0	52.5	48.5	44.0	35.1	24.7	21.6
		MCWB	15.5	18.0	18.2	22.6	29.7	36.2	49.7	47.0	42.8	34.2	24.0	21.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.8	29.4	29.9	32.2	34.0	50.4	63.1	57.2	52.3	39.4	31.4	29.7
		MCDB	28.4	29.7	30.3	32.6	34.5	55.6	69.0	59.7	53.8	40.0	31.5	30.0
	2%	WB	24.1	26.1	25.7	29.4	32.6	44.0	57.6	52.3	47.7	37.7	29.3	27.7
		MCDB	24.7	26.6	26.2	30.1	33.1	45.9	61.9	53.8	49.0	38.4	29.7	28.2
	5%	WB	20.6	22.6	22.3	26.2	31.5	39.4	53.4	49.5	45.3	35.9	27.0	25.1
		MCDB	21.3	23.1	22.7	26.8	32.0	40.3	56.3	51.1	46.0	36.6	27.7	25.6
	10%	WB	15.7	18.0	18.2	22.5	29.8	36.3	49.9	47.3	43.0	34.4	24.1	20.8
		MCDB	16.3	18.7	18.6	23.3	30.5	36.9	52.4	48.5	43.9	35.1	24.7	21.3
Mean Daily Temperature Range	5% DB	MDBR	13.3	12.9	13.4	13.9	8.4	4.9	7.9	5.7	4.6	6.2	11.6	12.5
		MCDBR	16.8	13.6	14.1	11.7	7.4	11.0	15.0	11.5	7.1	5.0	10.4	12.0
		MCWBR	16.6	13.7	14.2	11.4	7.4	9.3	10.8	9.2	6.4	5.1	10.5	11.7
	5% WB	MCDBR	16.9	13.9	14.2	11.8	7.2	11.0	14.6	10.1	7.0	4.8	10.3	12.2
		MCWBR	16.6	13.9	14.3	11.6	7.2	9.3	11.2	8.7	6.5	4.9	10.4	12.3
Clear-Sky Solar Irradiance	taub	N/A	0.203	0.277	0.300	0.324	0.328	0.362	0.344	0.305	0.350	N/A	N/A	
	taud	N/A	1.865	2.001	2.053	2.076	2.303	2.405	2.457	2.421	2.686	N/A	N/A	
	Ebn at Noon	0	100	201	250	261	268	252	241	207	101	0	0	
	Edh at Noon	0	9	24	34	39	32	29	23	17	7	0	0	
All-Sky Solar Radiation	RadAvg	0	48	469	1226	1760	1779	1428	851	376	78	0	0	
	RadStd	0	5	38	102	128	193	179	97	28	10	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page